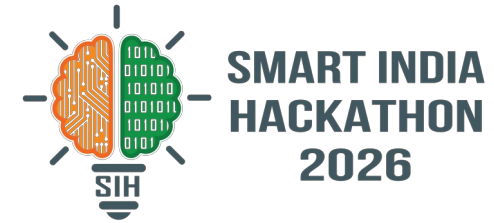


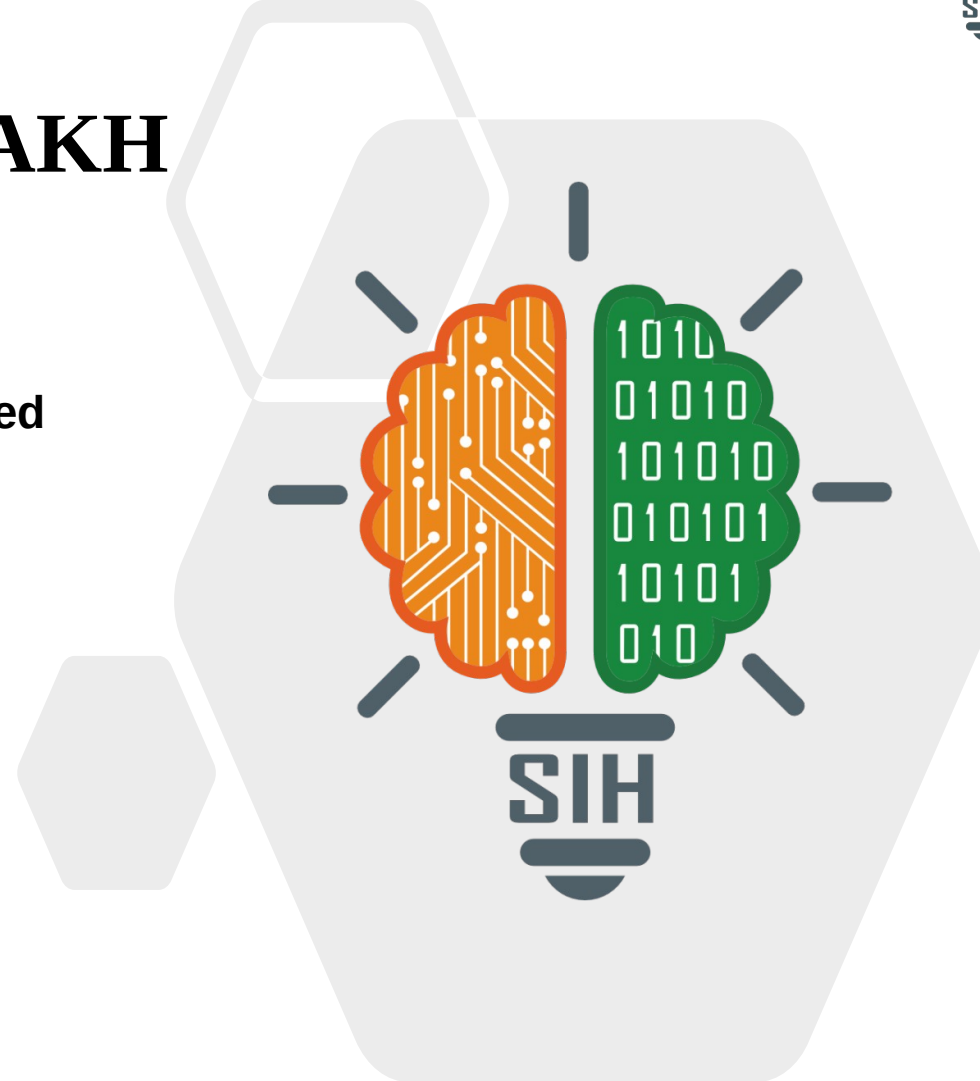
SMART INDIA HACKATHON 2026

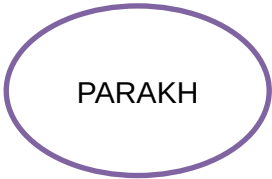


PARAKH

Problem Statement ID – 26103

- **Problem Statement Title - Use case on web-based integrated project-monitoring platform**
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - [To be provided]**
- **Registered Team Name - [To be provided]**





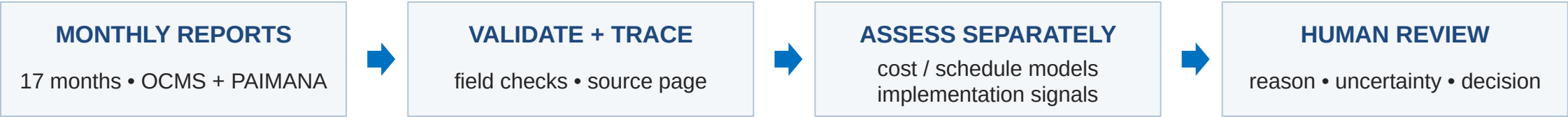
PARAKH: Evidence before alert

❖ A warning is useful only when its evidence can be trusted.



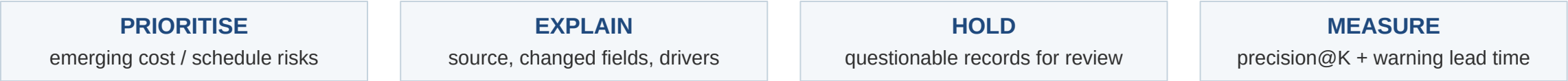
SOURCE: MoSPI PAIMANA FLASH REPORT • APRIL 2026 • TABLE 6

The target workflow for MoSPI and ministry monitoring teams compares each project’s monthly cost, schedule, expenditure and progress to surface emerging risks for review. A quality gate holds inconsistent records; each reviewed flag shows its source page, changed fields, model drivers and uncertainty, and lets an analyst record a decision.



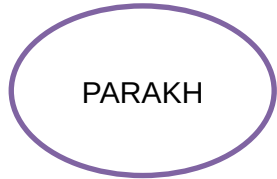
IF CHECKS FAIL → HOLD FOR REVIEW; NO SILENT PREDICTION

DESIGNED TO DELIVER

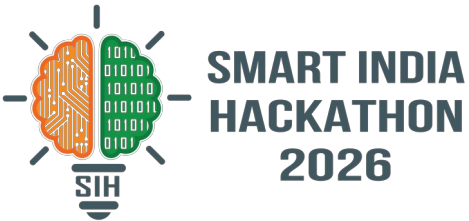


WORKING NOW • 17 report months, source/page references, quality status and a transparent score | NEXT • connect the gate to the analyst queue

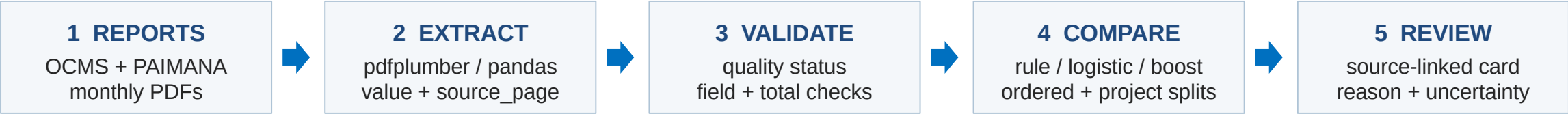
OPEN STACK • pdfplumber + pandas • scikit-learn • Next.js



TECHNICAL APPROACH



CURRENT PANEL • 17 reports (Mar 2025–Jul 2026) • 25,116 project-month rows • 3,950 project histories



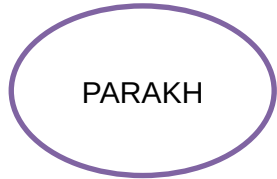
QUALITY CONTROL • Across 17 reports: 24,257 VERIFIED; 859 REVIEW_REQUIRED. Training uses VERIFIED rows. Flags are investigation cues, not confirmed errors.

ORDERED TIME-HOLDOUT • PR-AUC (rare-event ranking), not “accuracy”

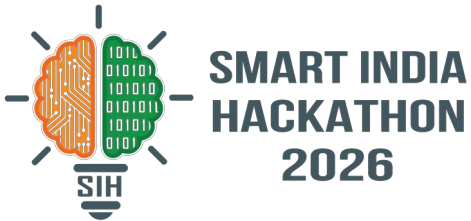
SCHEDULE CHANGE • APR–JUN 2026 (3 ORDERED MONTHS)	COST REVISION • JUNE 2026 ONLY (72 POSITIVES)
PR-AUC Gradient boost 0.376 Logistic 0.340 Rule 0.201	PR-AUC Logistic 0.131 Gradient boost 0.076 Rule 0.064
ROC-AUC Gradient boost 0.777 Logistic 0.762 Rule 0.566	ROC-AUC Logistic 0.672 Gradient boost 0.704 Rule 0.586
Mean across ordered tests; June test included 1,732 rows / 268 positive cases.	One ordered test month only: not decision-ready evidence.

EXPLAINABILITY • Show changed source values and permutation drivers (predictive signal, not cause). Next: CUF-only vs added-feature ablation and a human review disposition. Probabilities are uncalibrated; show ranks, not percentages.

NEXT VALIDATION • repeat time-forward checks | test top-K precision and lead time | add XGBoost / SHAP only if they improve validated results and explanation quality



FEASIBILITY AND VIABILITY



TECHNICAL

Existing Python, pandas, scikit-learn and Next.js stack; batch processing, no GPU requirement.

ECONOMIC

Open-source tooling; no paid model API. Estimate hosting and reviewer effort during pilot.

OPERATIONAL

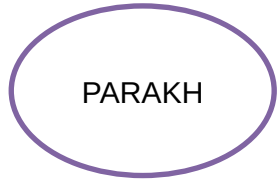
Monthly, analyst-led queue. No external alert dispatch; notifications follow roles and thresholds.

30-DAY BUILD PATH • TARGETS, NOT CLAIMED AS COMPLETE

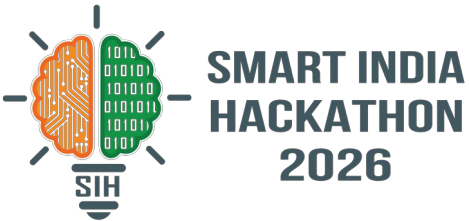
WEEK 1 Close source/reconciliation gaps; wire quality holds.	WEEK 2 Re-run ordered and project-disjoint benchmarks; compare CUF features.	WEEK 3 Build source-linked review cards and reviewer dispositions.	WEEK 4 Replay held-out months; set pilot precision@K and lead-time baselines.
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RISKS AND RESPONSES

Panel: 859 flagged rows	Flag patterns are not all extraction errors; preserve source pages and let analysts confirm or dismiss.
Cost evidence is thin	Only one ordered cost test month; withhold operational claims until repeated holdouts and calibration.
PDF layouts can change	Keep per-report schema/total checks; stop exports on hard failures, route warnings for review.
Alerts can overwhelm	Set top-K to reviewer capacity; measure precision, lead time and effort before notifications.



IMPACT AND BENEFITS



The goal is better review attention—not an unmeasured savings claim.

TODAY • MONTHLY STATUS REPORTING

Analysts can search current project records, but finding the next project worth reviewing still depends on evidence quality and reviewer attention.

PROPOSED • EVIDENCE-GATED REVIEW

A ranked, source-linked queue directs a reviewer to a change, the reason it was flagged, and any data limitation—then records the analyst’s disposition.

ANALOGY

Like an inspection triage board: it points to the project and evidence to check; the authorized official decides what to do.

PILOT SCORECARD • COMPARE WITH THE CURRENT REVIEW BASELINE

PRECISION@K

Useful flags within staff capacity

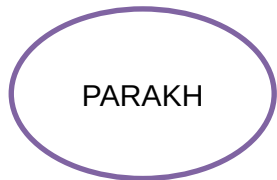
LEAD TIME

How early before a reported change

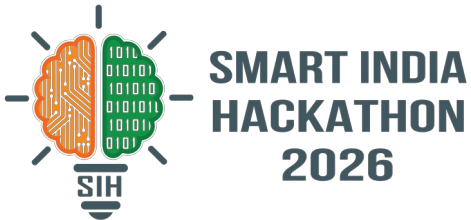
REVIEW EFFORT

Minutes and dispositions per case

No time saved, cost saved, delay reduction or environmental impact has yet been measured.



RESEARCH AND REFERENCES



Sources and prototype evidence

[MoSPI • PAIMANA Flash Report • April 2026](#)

Portfolio scale and Table 6: 1,981 projects, 17 ministries/departments, ₹42.78 lakh crore revised cost.

[MoSPI • PAIMANA public dashboard](#)

Official project-monitoring context and public portfolio view.

[SIH 2026 • Problem Statement 26103 • MoSPI / DIID](#)

Team-provided challenge brief: prediction, baseline comparison, CUF assessment and early-warning decision support.

[PARAKH • source and prototype artifacts](#)

17 monthly source PDFs; src/checker.py; src/model_train.py; data/processed/model_metrics.json and portfolio_kpis.json.

[Model evaluation • current saved run](#)

17 ordered report months; latest evaluation through June 2026. Cost results have one ordered test month; probabilities are uncalibrated.

The repository URL is taken from the configured Git remote; confirm it is publicly accessible before submission. Team name, Team ID and demo URL remain for the team to provide.